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LAB 1: REQUIREMENTS ENGINEERING & UML USE-CASE MODELLING

Problem Title: Customizable Subscription Box Scheduler

The Customizable Subscription Box Scheduler is a subscription-box portal that allows subscribers to customize their monthly product selections based on preference tags. Subscribers can swap items and pause or skip upcoming billing cycles within the permitted deadline. The system also supports fulfillment operations by allowing the Fulfillment Lead to generate fulfillment manifests and shipping labels for monthly subscription boxes.

Actor	**Description**
Subscriber	Customizes monthly boxes, manages preferences, and pauses or skips billing cycles.
Fulfillment Lead	Manages fulfillment manifests and shipping labels for monthly boxes.
Payment Gateway	External system that processes subscription billing.

Requirements Table:

Req ID	***Type***	***Description***	***Priority***	***Acceptance Criteria***	***Rationale***	***Comments***
FR-001	Functional	The system shall allow subscribers to swap monthly box items or pause deliveries up to 48 hours prior to the monthly	High	**Pass:** Selection/pause request is accepted before the 48-hour deadline and the fulfillment manifest is refreshed. **Fail:** Request is rejected	Core customization and subscription-management functionality.	Clear and testable.

		billing renewal date.		after the deadline.		
FR-002	Functional	The system shall allow subscribers to select monthly box items based on their preference tags.	High	**Pass:** When preference tags are selected, the system displays matching eligible products and allows selection. **Fail:** Products unrelated to the selected tags are incorrectly presented as matches.	Enables personalized subscription boxes.	Specifies how preferences affect selection.
FR-003	Functional	The system shall allow subscribers to skip an upcoming billing cycle before the cycle's modification deadline.	High	**Pass:** A valid skip request marks the upcoming cycle as skipped and prevents billing for that cycle. **Fail:** The cycle remains active or billing occurs after a successful skip request.	Required because skipping cycles is explicitly supported by the scenario.	Deadline enforcement should be verified.
FR-004	Functional	The system shall allow the Fulfillment	High	**Pass:** The generated manifest contains all	Supports monthly	Manifest must reflect

		Lead to generate a fulfillment manifest containing the finalized product selections for the upcoming billing cycle.		finalized boxes and their selected items. **Fail:** Any finalized box is missing or contains incorrect selections.	fulfillment operations.	finalized selections.
FR-005	Functional	The system shall notify subscribers when their monthly box customization is successfully saved or when a customization request is rejected.	Medium	**Pass:** A notification is generated within 1 minute of the request outcome. **Fail:** No notification is generated or the notification reports an incorrect status.	Provides clear confirmation of subscriber actions.	Notification timing is measurable.
NFR-001	Nonfunctional — Performance & Security	The fulfillment manifest generator shall export shipping labels for 10,000 monthly boxes in under 1 minute while maintaining the required	High	**Pass:** Benchmarking confirms that 10,000 labels are generated in less than 60 seconds during simulated peak load without unauthorized data exposure.	Ensures efficient and secure peak-load fulfillment.	Performance target is measurable.

		security standards.		**Fail:** Generation takes 60 seconds or longer or protected data is exposed.		
NFR-002	Nonfunctional — Availability	The system shall maintain 99.5% monthly availability, excluding scheduled maintenance.	Medium	**Pass:** Monitoring records at least 99.5% availability during the measurement period. **Fail:** Availability falls below 99.5%.	Ensures reliable access for subscribers and fulfillment staff.	Availability target is measurable.

ACTORS:

Actor	Description
Subscriber	A user who customizes monthly box items, manages preference tags, and pauses or skips billing cycles.
Fulfillment Lead	Manages fulfillment manifests and generates shipping labels for monthly subscription boxes.
Payment Gateway	An external system that processes monthly subscription billing.

USE CASES:

Use Case ID	Use Case	Primary Actor
UC-01	Customize Monthly Box	Subscriber

Use Case ID	Use Case	Primary Actor
UC-02	Manage Preference Tags	Subscriber
UC-03	Swap Box Items	Subscriber
UC-04	Pause/Skip Billing Cycle	Subscriber
UC-05	Process Monthly Billing	Payment Gateway
UC-06	Generate Fulfillment Manifest	Fulfillment Lead
UC-07	Generate Shipping Labels	Fulfillment Lead

The use cases represent the major interactions between the actors and the Customizable Subscription Box Scheduler. Subscribers primarily manage and customize their monthly subscriptions, while the Fulfillment Lead manages the fulfillment process. The Payment Gateway interacts with the system to process monthly billing.

UML Use-case diagram:

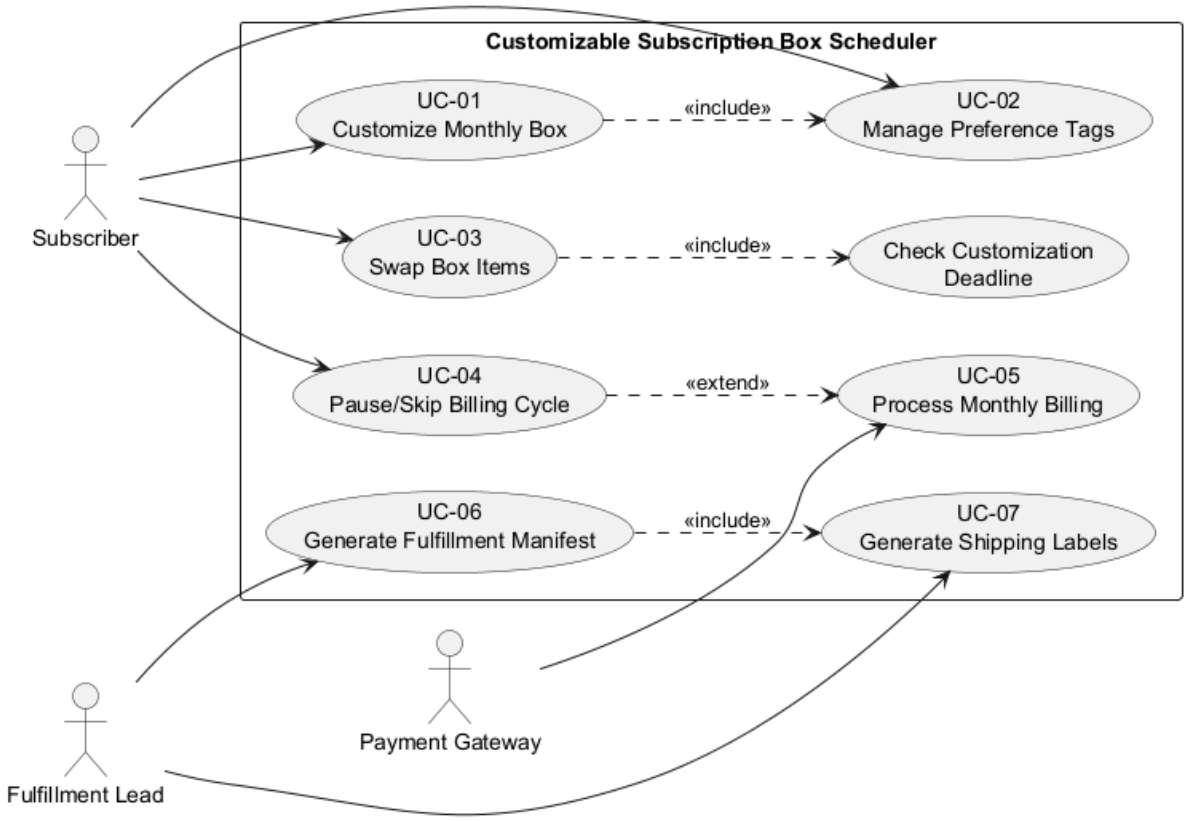


Figure 1: UML Use-Case Diagram for the Customizable Subscription Box Scheduler

USE-CASE Flow:

Use Case: UC-01 — Customize Monthly Box

Primary Actor: Subscriber
Goal: Customize the upcoming monthly subscription box.

Main Success Scenario

1. The Subscriber logs into the subscription box portal.
2. The Subscriber selects **"Customize Monthly Box."**
3. The system displays the Subscriber's preference tags and eligible products.
4. The Subscriber selects or modifies preference tags.
5. The system displays products matching the selected preferences.
6. The Subscriber selects the desired products for the upcoming box.
7. The system verifies that the customization request is made at least **48 hours before the monthly billing renewal deadline**.
8. The system saves the updated box selection.
9. The system refreshes the fulfillment information for the Subscriber's box.
10. The system displays a confirmation message indicating that the monthly box has been successfully customized.
11. The use case ends successfully.

Alternate Flow A — Customization Deadline Passed

Condition: The Subscriber attempts to customize the box less than 48 hours before the monthly billing renewal date.

1. The system determines that fewer than 48 hours remain before the billing renewal date.
2. The system prevents the Subscriber from modifying the upcoming box.
3. The system displays a message stating that customization is locked because the deadline has passed.
4. The existing box selection remains unchanged.
5. The system does not update the fulfillment information.

6. The use case ends without making any changes.

Alternate Flow B — Selected Product Unavailable

Condition: A selected product is no longer available.

1. The system detects that one or more selected products are unavailable.
2. The system informs the Subscriber which product(s) cannot be selected.
3. The system asks the Subscriber to choose alternative products.
4. The Subscriber selects available replacement products.
5. The system continues from **Step 7 of the Main Success Scenario**.

Postcondition

If the customization is successful, the Subscriber's monthly box contains the newly selected products and the corresponding fulfillment information is updated. If the customization deadline has passed, the existing box selection remains unchanged.

Conclusion:

The requirements and use-case model define the major functional and nonfunctional aspects of the Customizable Subscription Box Scheduler. The use-case flow demonstrates the successful customization of a monthly box and handles important alternate conditions, including a missed customization deadline and an unavailable product.